

Quantifying Ocular Confounds in EEG Decoding Without an Eye Tracker

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Abstract—Eye movements, blinks, and the corneo-retinal standing potential leak into scalp electroencephalography (EEG) and co-vary with task structure, so a classifier that claims to decode a cognitive state may instead be partly reading the eyes. Existing defenses either remove ocular signal or model it within a single eye-tracked recording; none provides a standardized, montage-matched audit returning a calibrated ocular-predictability estimate for completed EEG decoding datasets that lack an eye-tracker. We define and validate an auditing instrument. A frozen linear EEG-to-gaze readout, trained where synchronized gaze exists, induces a locked ocular subspace (the orthogonal projection onto the span of its weights). The auditor operates on all available channels: we report a full 64-channel montage and, as an explicitly specified subset, the 30-channel ERP-CORE set. The Ocular Predictability Index (OPI) is the cross-validated area under the curve (AUC) of a paradigm-matched decoder applied to that projection, minus a within-subject label-permutation null. The instrument ships with a three-tier reliability gate that returns “inconclusive” when gaze cannot be reconstructed in the target montage, a measured two-sided bias envelope, and a computable neural-leakage diagnostic. We validate against ground truth in a physically grounded EEG-plus-gaze simulator and demonstrate the audit logic on real 64-channel EEG that lacks eye tracking (PhysioNet eegmmidb). In simulation, the OPI is 0.43 on saccade-coupled ocular data and 0.20 on a condition-correlated eye-blink artifact, versus at most 0.03 on every negative control, rising monotonically with the artifact-label coupling. A lateralized neural source decodable from full EEG (AUC 0.79) is not decodable from the ocular projection (OPI 0.007, not significant). Neural leakage tracks the principal-angle overlap between the cognitive topography and the ocular subspace (Spearman correlation 0.95). A frozen, montage-matched ocular-subspace auditor thus estimates how much of a decoding result could be ocular: a calibrated number where montage and source conditions are validated, and “inconclusive” when reliability is insufficient, enabling ocular-leakage audits of EEG decoding results recorded without an eye-tracker.

Index Terms—EEG decoding, ocular artifact, eye movements, construct validity, confound auditing, brain-computer interface, subspace projection

I. Introduction

A decoding result is only as trustworthy as the signal it rests on. In EEG, a recurring threat to construct validity is that a classifier credited with reading a cognitive state (attention, working-memory load, target detection, the meaning of a read word) is partly reading the eyes. The eye is an electrical dipole, cornea-positive and retina-negative, with a standing potential of roughly 0.4–1.0 mV [1]; its rotations, together with blinks and the broadband saccadic spike potential at saccade onset [2], project to frontal and periocular electrodes with amplitudes that dwarf cortical signals. Crucially, the field of an eye movement depends on each electrode’s position relative to the eyes: its sign reverses across the midline (a dipole) and its magnitude falls with distance, and a blink imposes a different, symmetric pattern, so the contamination an electrode sees is montage- and artifact-specific. Because where and when a person looks is rarely independent of the experimental condition, this ocular activity is condition-correlated: it can be decoded in lockstep with the cognitive label and inflate apparent accuracy. The danger is the machine-learning failure mode of shortcut learning: exploiting an incidental correlate rather than the intended construct [3].

The standard defenses are incomplete. Independent component analysis and regression remove ocular components [4], [5], but removal is imperfect and leaves a residual whose decodable content is unknown; overweighting the saccadic spike before ICA improves this but still targets removal, not measurement [6], [7]. Deconvolution frameworks rigorously model ocular and overlap effects, but in practice within a single, carefully co-registered, eye-tracked recording [8], [9]. None of these answers the question an author or a reviewer actually faces about a finished result: what fraction of this decodable signal could be ocular? This is especially true for the thousands of datasets recorded with no eye-tracker at all.

The concern is not hypothetical. Mostert et al. [10] showed that a visual working-memory “representation” could be partly explained by systematic eye movements, and constructed a bespoke, within-experiment control: if the target variable cannot be decoded from simultaneously recorded gaze, the neural result is safer. Rotaru et al. [11] found that EEG decoders of the spatial focus of auditory attention overfit to eye-gaze and per-trial fingerprints, dropping to chance once gaze was controlled. More broadly, the field increasingly rewards auditing its methods and benchmarks for inflated inference [12]–[15]. What is missing, to our knowledge, is a transferable instrument: a frozen, montage-matched ocular-subspace probe that can be pointed at an arbitrary EEG decoding dataset (including one with no gaze recording) and that returns a calibrated number together with an honest statement of when it cannot be trusted.

This paper defines and validates that instrument. Our contributions are: (i) a locked, linear ocular-subspace operator and an Ocular Predictability Index (OPI) with a within-subject permutation null and a full-EEG ceiling; (ii) a three-tier reliability gate that returns “inconclusive” rather than a false clean bill of health; (iii) a measured two-sided bias envelope, so the index can neither silently hide contamination nor over-blame the eyes; and (iv) a computable neural-leakage diagnostic: the one residual failure mode, that the projection absorbs genuine neural signal, is shown to be governed by the principal angle between the cognitive topography and the ocular subspace, converting a reviewer’s objection into a number. We validate all four against ground truth in a physically grounded simulator and demonstrate the instrument on real 64-channel EEG.

II. Methods

A. Design and rationale

The instrument is deliberately linear. Although a deep network is one way to learn an EEG-to-gaze readout where synchronized eye-tracking is available [16], [17], the object that is frozen and shipped is a linear readout and the orthogonal projection onto the span of its weights. We therefore implement and validate exactly that object. Because ground truth (knowing how much of a decodable signal is truly ocular) is available only when gaze is recorded or simulated, the primary validation uses a forward-model simulator with known sources, complemented by a real-EEG negative control.

The auditor uses every channel a dataset provides; dimensionality is never silently reduced. We report a full 64-channel 10–10 montage as primary and treat the 30-channel ERP-CORE set as an explicitly specified subset of it, used only to characterise how channel count and near-eye frontal coverage affect the auditor (the reliability gate, below). Validation draws on more than one dataset: EEGEyeNet (hundreds of subjects with 128-channel EEG and synchronized eye-tracking) is the source paradigm and montage family from which the gaze readout would

be trained [16]; ERP CORE supplies the 30-channel target montage [18]; PhysioNet eegmldb supplies real 64-channel EEG [19]; and ZuCo provides a same-cap-family reading corpus for the near-fixation regime [17]. Source amplitudes are taken from the electrooculography and saccade literature.

B. The locked ocular-subspace operator

Let $\mathbf{X}_{\text{src}} \in \mathbb{R}^{n \times p}$ be source EEG (trials $\times$ channels) for which ocular ground truth $\mathbf{Y}_{\text{src}} \in \mathbb{R}^{n \times q}$ is available (q outputs: horizontal and vertical gaze, a saccade channel, and a blink indicator). A ridge readout $\mathbf{W} = \arg \min_{\mathbf{W}} \|\mathbf{Y}_{\text{src}} - \mathbf{X}_{\text{src}} \mathbf{W}\|_F^2 + \lambda \|\mathbf{W}\|_F^2$ maps EEG to ocular state. The ocular subspace is the column space of $\mathbf{W}$, and the locked operator is the orthogonal projection onto it,

$$\mathbf{P} = \mathbf{U}\mathbf{U}^\top, \quad \mathbf{U} = \text{left singular vectors of } \mathbf{W}, \quad (1)$$

a symmetric idempotent $p \times p$ matrix ($\mathbf{P}^2 = \mathbf{P}$). Projecting target EEG, $\mathbf{X} \mapsto \mathbf{X}\mathbf{P}$, retains only the component lying in the ocular readout’s span. The gaze readout is trained only on source data with synchronized gaze, and the operator is fixed by (1). For a target with a different montage, the readout is re-fit on source data restricted to that montage (montage-matched, no periocular interpolation); the operator is thus frozen per montage and applied unchanged to every target, never altered by target data or target labels.

C. The Ocular Predictability Index

A paradigm-matched cognitive decoder (a ridge linear classifier) is trained and tested under strictly subject-wise cross-validation on the projected data $\mathbf{X}\mathbf{P}$, yielding a macro-averaged AUC. The OPI subtracts a within-subject label-permutation null (labels shuffled within each subject, $B = 200$):

$$\text{OPI} = \text{AUC}_{\text{proj}} - \overline{\text{AUC}}_{\text{null}}, \quad (2)$$

and is reported alongside the full-EEG AUC as a ceiling and a permutation p -value. The ocular-subspace operator is learned without target cognitive labels; the downstream audit decoder uses those labels only to quantify how much label information survives in the ocular projection.

D. Reliability gate and bias envelope

Reliability is the subject-wise cross-validated gaze-reconstruction R^2 of the readout restricted to the target montage, computed on source data with synchronized gaze (leave-one-subject-out over source subjects), never on the target. Two montages are matched when they share the channels the readout relies on, chiefly near-eye frontal coverage, so this is a montage-level competence score rather than a subject-level one. Transfer to an unseen target rests on one stated assumption: the linear ocular-to-scalp map is set by electrode geometry and volume conduction from a common ocular generator, so a readout that reconstructs gaze across held-out source subjects on a montage should do so for a new subject on the same

montage. A no-eye-tracker target thus inherits the score of its matched source montage; where none matches (for example, too few frontal channels to recover gaze), the gate returns inconclusive rather than a number. Thresholds are derived empirically from the reliability–bias relationship and partition results into confirmatory, exploratory, and inconclusive tiers. The bias envelope is measured, not assumed: where ground-truth gaze exists, we compare the auditor’s OPI to the OPI computed directly from true gaze across the saccade-amplitude range and report the signed deviation as a two-sided margin.

E. The neural-leakage diagnostic

The instrument’s central risk is that the projection absorbs genuine neural activity that happens to be gaze-correlated, inflating the index. We bound this. For a target whose cognitive decoder has contrast topography $\mathbf{t}$, define the overlap

$$s = \frac{\|\mathbf{P}\mathbf{t}\|}{\|\mathbf{t}\|} = \cos \theta, \quad (3)$$

the cosine of the principal angle θ between $\mathbf{t}$ and the ocular subspace. Here s is a linear topographic-overlap diagnostic: it gives the fraction of the (linear) contrast topography lying in the ocular subspace, and is 0 when the two are orthogonal. Turning s into decodable leakage further depends on the decoder, noise, and signal amplitude, so rather than a closed-form bound we report s per audit and show empirically that leakage is monotone in s .

F. Grounded EEG+gaze simulator

Ground-truth validation uses a forward-model simulator on a full 64-channel 10–10 montage; the 30 ERP-CORE electrodes [18] are an explicitly labelled subset of these 64, so the auditor is exercised on all 64 channels and the 30-channel case is a deliberately studied restriction, never a silent reduction. Per-trial channel patterns are sums of physically grounded source topographies: a horizontal corneo-retinal field ($\approx 16 \mu\text{V}$ per degree), a vertical/blink frontopolar field, a focal periocular saccadic spike scaling with saccade size [2], an $\approx 80 \mu\text{V}$ frontopolar blink, and a lateralized posterior N2pc-like neural component ($\approx 1.8 \mu\text{V}$) [20], on spatially correlated background noise. An artifact is coupled to the cognitive label with tunable strength, implementing the condition-correlated premise. Scenarios include saccade-coupled ocular and blink-coupled ocular (two positive controls testing distinct artifact types), covert attention (lateralized posterior neural source, central gaze), focal motor imagery, label shuffle, and a neutral baseline.

For reproducibility: each trial is a single channel-space feature vector, and each source is a fixed Gaussian scalp topography scaled by a per-trial amplitude, added to a shared low-rank smooth field plus per-channel white noise ($\approx 3 \mu\text{V}$ each). The coupling ρ sets the probability that gaze (or, for blinks, the blink rate) follows the label. The readout is trained on 20 source subjects $\times$ 200

trials at 8° saccades; each evaluation cell is 12 subjects $\times$ 120 trials over 12 fixed-seed studies (error bars are their SD), with ridge $\lambda=1$ fixed (not tuned in-fold), leave-one-subject-out macro AUC, and a $B=200$ within-subject label-permutation null. Complete generative equations and parameters are in the released code.

G. Real data

We use PhysioNet eegmimdb (all 64 channels, BCI2000, 160 Hz, EDF+; 8 subjects) [19], [21]; no channels are dropped. Because eegmimdb has no eye-tracker, the ocular subspace is estimated from the eyes-open baseline run of four source subjects (a real-EEG analogue of a frozen gaze-trained auditor) and applied to four held-out target subjects’ motor runs. We audit two contrasts: left-versus-right imagined movement (no differential gaze expected; the clean negative control) and rest-versus-movement.

H. Implementation

The auditor, simulator, and analyses are implemented in R (CPU-only). AUC is computed by the rank statistic and cross-checked against the pROC package [22] to $< 10^{-6}$. The trained auditor is small and CPU-deployable. The deployment vision, applying the frozen instrument across public decoding benchmarks (e.g., [18], [23]–[25], accessed via MOABB [26], [27]), is discussed in Section IV.

III. Results

A. The operator behaves as specified

Unit tests confirm the rank-based AUC matches pROC to $< 10^{-6}$; the projector of (1) is symmetric, idempotent, and correct-rank; and end-to-end the OPI is large when the cognitive signal lies in the ocular subspace and null when it lies outside. The trained auditors had ocular-subspace rank 4 on the simulator and rank 2 on real eegmimdb.

B. Graded sensitivity and specificity (H3)

Across 12 independent simulated studies per cell on the full 64-channel montage, the OPI cleanly separates ocular leakage from neural signal (Table I). The two positive controls test distinct artifact types: saccade-coupled ocular gives OPI = 0.429 with projected AUC equal to the full-EEG ceiling (0.929), and a condition-correlated eye-blink artifact gives OPI = 0.198 (projected AUC 0.697), both significant in 100% of studies. The auditor therefore detects blink contamination as well as saccade contamination. Every negative control is near-null in magnitude: motor imagery 0.031, label-shuffle 0.007, covert attention 0.007, neutral -0.003 ; their envelope sets an empirical calibrated neural-leakage margin of ≈ 0.05 that the index must clear beyond the bare permutation null. Motor imagery reaches significance in 42% of studies—a tight within-subject null plus a small sensorimotor–ocular overlap—yet its OPI of 0.031 sits below this margin, which is why the decision rule is the margin, not bare

TABLE I

H3 sensitivity battery: the OPI separates ocular leakage from neural signal (mean of 12 simulated studies; full 64-channel montage).

Scenario	OPI	proj	ceiling	% sig
Saccade-coupled ocular ^a	0.429	0.929	0.930	100
Blink-coupled ocular ^a	0.198	0.697	0.781	100
Motor imagery	0.031	0.531	0.842	42
Covert attention ^b	0.007	0.507	0.785	17
Label shuffle	0.007	0.507	0.505	0
Neutral	−0.003	0.496	0.635	0

proj = AUC on the ocular projection; ceiling = full-EEG AUC; % sig = studies with permutation $p < 0.05$.

^a Positive control (two distinct artifact types).

^b Neural-leakage test: decodable from full EEG, not from the ocular subspace.

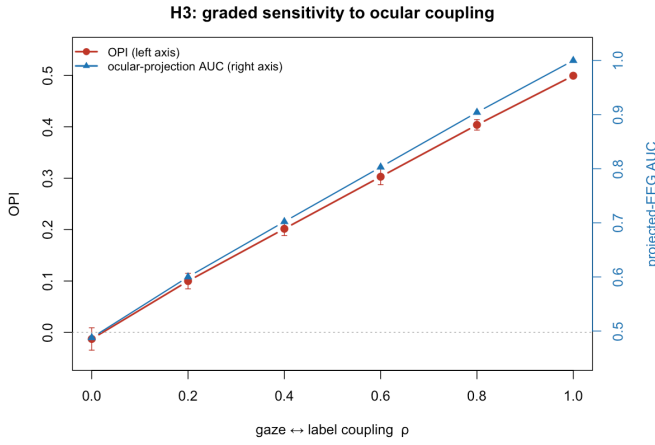


Fig. 1. Graded sensitivity: OPI (red, left axis) and ocular-projection AUC (blue, right axis) rise monotonically with the gaze-label coupling.

significance. Decisively, the covert-attention contrast is decodable from full EEG (ceiling AUC 0.79) yet not from the ocular projection (OPI 0.007, $p = 0.48$): the auditor does not absorb lateralized posterior neural signal.

The index is a graded meter, not a binary flag: as the gaze-label coupling ρ increases from 0 to 1, the OPI rises monotonically and near-linearly from -0.014 to 0.500 , with the projected AUC tracking from chance to 1.0 (Fig. 1).

C. The bias envelope is conservative

Comparing the auditor’s OPI to the OPI from ground-truth gaze across the saccade-amplitude range (Table II), the bias is within ± 0.01 from 0.5° upward on both montages. Only at the fixational extreme (0.25°) does it grow, to -0.034 on the full 64-channel montage and -0.056 on the 30-channel subset, and in both cases the auditor understates the ocular contribution, the conservative direction. No over-statement was observed. The breach magnitude

TABLE II

Reliability (R^2), bias, and trust tier across saccade amplitude and montage.

Amp. (deg)	R^2	bias	tier
Full montage (64 ch)			
0.25	0.664	−0.034	confirmatory
1.0	0.899	0.000	confirmatory
12.0	0.755	0.000	confirmatory
ERP-CORE subset (30 ch)			
0.25	0.497	−0.056	exploratory
1.0	0.775	0.000	confirmatory
12.0	0.679	0.000	confirmatory

Thresholds (empirical): confirmatory $R^2 \geq 0.664$, exploratory $R^2 \geq 0.497$. Representative amplitudes shown; full grid in released tables.

is measured and folded into the trust tiers as a two-sided margin.

D. The reliability gate has teeth

The empirically calibrated thresholds are $R^2 = 0.664$ (confirmatory) and $R^2 = 0.497$ (exploratory). Tiering the amplitude $\times$ montage grid (Table II), the full 64-channel montage is confirmatory at every amplitude, while the 30-channel subset falls to exploratory at the fixational extreme, precisely the regime (sparse montage, sub-degree movement) where a silent failure would be most dangerous.

E. Neural leakage tracks a measurable overlap angle

Sweeping a lateralized neural source from posterior to frontal with no ocular coupling (so any positive OPI is pure leakage), the leakage is monotone in the overlap s of (3) (Fig. 2). It rises monotonically with s (Spearman $\rho = 0.95$ across the swept range), from $+0.006$ at near-orthogonality ($s \approx 0.02$, the posterior-N2pc regime) to $+0.12$ – 0.14 when the source is frontal ($s \approx 0.6$ – 0.8). The headline posterior case is thus low-leakage under the linear model, and any target with a frontal cognitive topography is flagged by a large s for the leakage-corrected envelope.

F. A headline-analog audit is gated honestly

To emulate the deployment case we audited a simulated ERP-CORE-like N2pc target on the sparse montage carrying a sub- 0.2° condition-correlated ocular residual of the kind that survives strict eye-movement rejection. The OPI was positive and significant (0.150, ceiling AUC 0.777, $p < 0.005$), but the reliability gate returned “inconclusive” ($R^2 = 0.429$, below the exploratory threshold). The instrument withholds certification of a positive, significant index because the montage and regime cannot reconstruct gaze well enough to trust it: the intended “inconclusive by design” safeguard, not a false alarm.

G. Real-EEG negative control

On real 64-channel eegmmidb with the ocular subspace frozen from independent subjects’ eyes-open baseline (Table III), the left-versus-right imagery contrast, where the

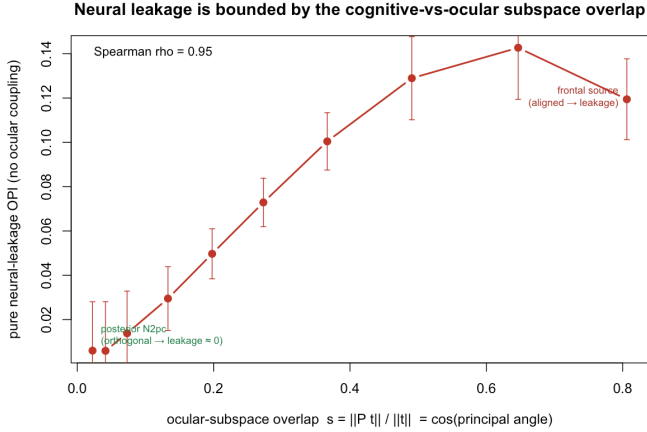


Fig. 2. Neural leakage versus the cognitive-vs-ocular subspace overlap $s = \cos \theta$. Pure neural-leakage OPI rises monotonically with s ($\rho = 0.95$) and is ≈ 0 for a posterior (orthogonal) source.

TABLE III

Real-EEG audit (eegmmidb; ocular subspace frozen from source subjects’ eyes-open baseline; subject-wise evaluation).

Contrast	ceiling	proj	OPI	p
Left vs. right imagery ^a	0.601	0.501	-0.023	0.998
Rest vs. move ^b	0.672	0.615	+0.114	0.002

Subject-wise macro-averaging over few subjects raises the permutation-null AUC slightly above 0.5 (here ≈ 0.52); a negative OPI means projected AUC below that null.

^a Clean negative control (no differential gaze): not decodable from the ocular subspace.

^b Cautionary: a no-eye-tracker channel proxy retains frontal neural signal.

eyes have no reason to differ, is weakly decodable from full EEG (AUC 0.601) but not from the ocular subspace (OPI -0.023, $p = 0.998$): a clean negative control on real data. The rest-versus-move contrast is included as a cautionary case: a frontal-channel-proxy subspace (the best obtainable without an eye-tracker) retains frontal neural signal it cannot separate from ocular activity (OPI +0.114, $p = 0.002$). This is the empirical argument for the instrument’s design: the auditor must be trained on synchronized gaze, which isolates the ocular subspace from frontal neural activity, rather than on channel proxies.

IV. Discussion

We have defined a frozen, montage-matched ocular-subspace auditor and shown, against ground truth, that it does what an audit instrument must: it is sensitive to more than one artifact type (OPI 0.43 for saccade-coupled and 0.20 for blink-coupled contamination), specific (near-null on neural and shuffled controls), graded (monotone in the coupling), conservative (a measured, signed bias envelope that errs toward under-stating), self-aware (a reliability gate that returns “inconclusive”), and, critically, honest about its one residual failure mode, neural leakage, which

we render as a computable number, the principal-angle overlap s . Throughout, the auditor uses all 64 channels; the 30-channel ERP-CORE case is reported as an explicit subset to quantify how channel count affects reliability, not as a hidden reduction.

Relation to prior work. The instrument generalizes the within-experiment control of Mostert et al. [10] and the single-domain diagnosis of Rotaru et al. [11] into a transferable meter that requires no eye-tracker on the target, complementing rather than replacing ocular removal [4], [6] and within-recording deconvolution [8]. Where confound-quantification in machine learning probes a measured nuisance variable [15], here the nuisance (gaze) is physically recovered even when it was never recorded in the target, and the back-door through which it could mislead (neural leakage) is diagnosed by the overlap of (3). This positions the work in the now-active genre of auditing the field’s results for inflated inference [12]–[14].

Toward a field-wide audit. Because the instrument is portable and CPU-cheap, the natural deployment is an ocular-leakage audit across widely used public benchmarks (ERP, P300 spellers, rapid visual streams, motor imagery [18], [23], [24], [28]), accessed uniformly via MOABB [26] and each paired with a gaze- or EOG-bearing anchor so that verdicts are ground-truthable. The competence boundary makes such a leaderboard honest: a benchmark on which gaze cannot be reconstructed is reported “inconclusive,” not falsely cleared.

A. Limitations

The ground-truth validation is necessarily simulation-based, because measuring bias requires knowing the true ocular contribution; the simulator is grounded in measured physiology but is a simplification (notably, spatial features rather than full spatiotemporal windows, and a linear generative structure aligned with the linear operator). The overlap score s bounds only the linear topographic overlap, and its monotonic relationship to decodable leakage is shown empirically under the linear-subspace model; both should be tested across real cognitive topographies. Real synchronized-gaze datasets such as EEGEyeNet [16] and ZuCo [17], [29] are the appropriate anchors for training the readout and validating verdicts, but their training paradigms are cued large saccades, so the sub-degree fixational regime must be anchored on reading microsaccades rather than assumed. The real-EEG demonstration used a channel-proxy subspace because eegmmidb has no eye-tracker; it therefore validates the negative-control logic and the design rationale, not a gaze-trained transfer. Finally, the index measures only the ocular subspace; it will not detect non-ocular confounds such as repeated-stimulus fingerprints.

V. Conclusion

How much of an EEG decoding result could be the eyes rather than the mind is, for most published datasets,

an unanswerable question because no eye-tracker was present. A frozen, montage-matched ocular-subspace auditor makes it answerable: it converts the worry into a calibrated, two-sided Ocular Predictability Index with a self-declared competence boundary and a computable neural-leakage diagnostic, validated here against ground truth and demonstrated on real EEG. The instrument is small, CPU-runnable, and released, and it is built to be pointed at the field’s benchmarks, reporting “inconclusive” wherever it honestly cannot see.

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